COMPUTER ENGINEERING

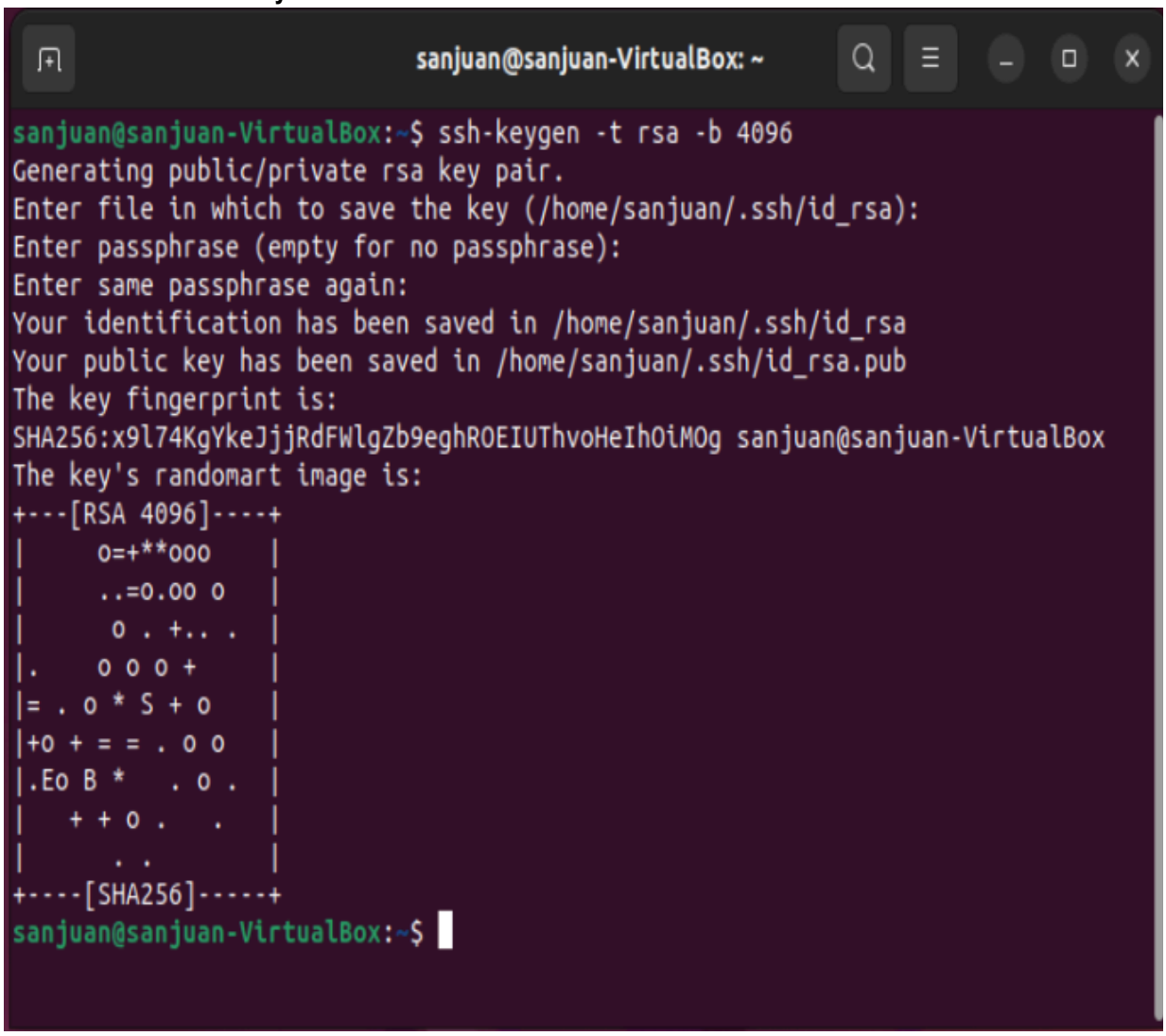
QUEZON CITY

COURSE CODE/TITLE:	CPE212	**SECTION:**	CPE31S2
DATE PERFORMED:	07/30/26	**DATE SUBMITTED:**	07/30/26
NAME:	San Juan, Ivan Wayne	**INSTRUCTOR:**	Engr. Jimlord Quejado

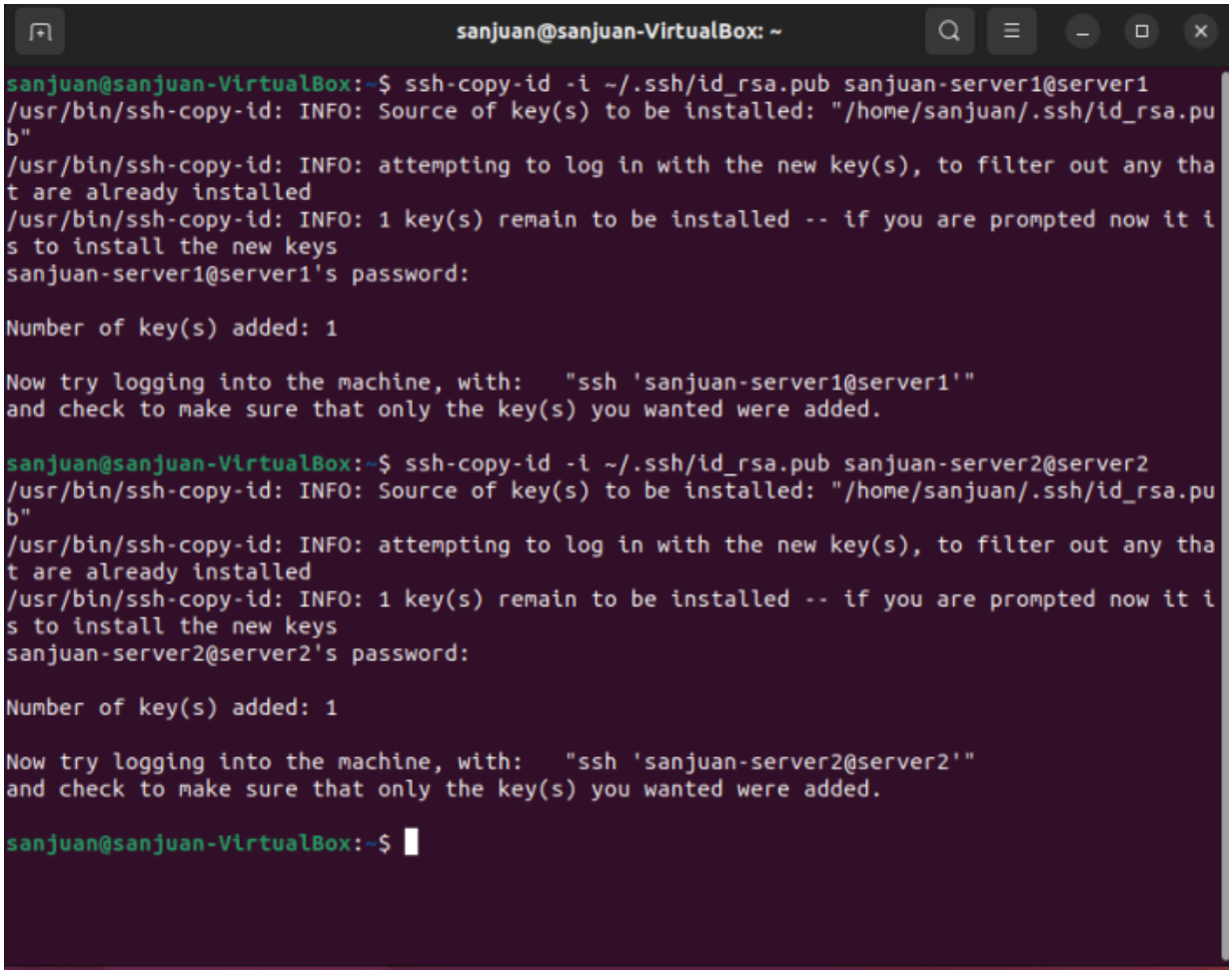
ACTIVITY NO. 2.1 **SSH Key-Based Authentication and GIT Setup**

1. PROCEDURE OUTPUT

Task 1: Create an SSH Key Pair for User Authentication



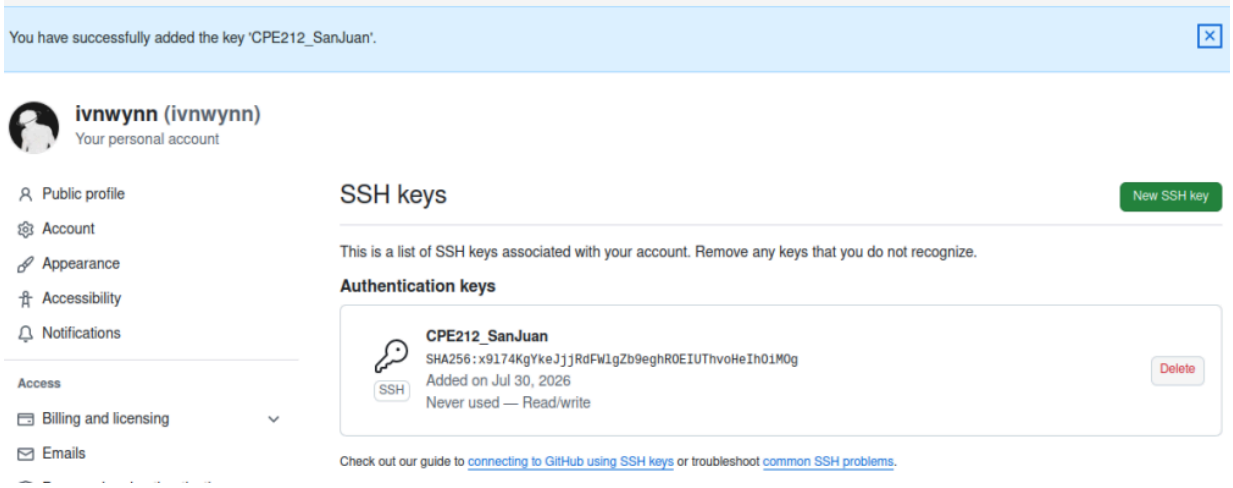
Task 2: Copying the Public Key to the remote servers



Task 3: Set up the Git Repository

```
sanjuan@sanjuan-VirtualBox:~$ sudo apt install git
[sudo] password for sanjuan:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libsmbios-c2 linux-headers-5.19.0-32-generic
  linux-hwe-5.19-headers-5.19.0-32 linux-image-5.19.0-32-generic
  linux-modules-5.19.0-32-generic linux-modules-extra-5.19.0-32-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  git-man liberror-perl
Suggested packages:
  git-daemon-run | git-daemon-sysvinit git-doc git-email git-gui gitk gitweb git-cvs
  git-mediawiki git-svn
The following NEW packages will be installed:
  git git-man liberror-perl
0 upgraded, 3 newly installed, 0 to remove and 0 not upgraded.
Need to get 4,154 kB of archives.
After this operation, 21.0 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://ph.archive.ubuntu.com/ubuntu jammy/main amd64 liberror-perl all 0.17029-1 [26.
5 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 git-man all 1:2.34.1-1ub
untu1.17 [954 kB]
Get:3 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 git amd64 1:2.34.1-1ubun
tu1.17 [3,174 kB]
Fetched 4,154 kB in 7s (596 kB/s)
```

```
Processing triggers for man-db (2.10.2-1) ...
sanjuan@sanjuan-VirtualBox:~$ which git
/usr/bin/git
sanjuan@sanjuan-VirtualBox:~$
```



```
juan@sanjuan-VirtualBox
sanjuan@sanjuan-VirtualBox:~$ git clone git@github.com:ivnwynn/CPE212_SanJuan.git
Cloning into 'CPE212_SanJuan'...
The authenticity of host 'github.com (20.205.243.166)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
sanjuan@sanjuan-VirtualBox:~$
```

```
sanjuan@sanjuan-VirtualBox:~$ cd CPE212_SanJuan
sanjuan@sanjuan-VirtualBox:~/CPE212_SanJuan$ ls
README.md
```

```
sanjuan@sanjuan-VirtualBox:~/CPE212_SanJuan$ git config -global user.name San Juan
error: did you mean `--global` (with two dashes)?
sanjuan@sanjuan-VirtualBox:~/CPE212_SanJuan$ git config --global user.name SanJuan
sanjuan@sanjuan-VirtualBox:~/CPE212_SanJuan$ git config --global user.email qiwvsanjuan@tip
.edu.ph
sanjuan@sanjuan-VirtualBox:~/CPE212_SanJuan$ cat ~/.gitconfig
[user]
        name = SanJuan
        email = qiwvsanjuan@tip.edu.ph
sanjuan@sanjuan-VirtualBox:~/CPE212_SanJuan$
```

```
sanjuan@sanjuan-VirtualBox:~/CPE212_SanJuan$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
sanjuan@sanjuan-VirtualBox:~/CPE212_SanJuan$ git add README.md
sanjuan@sanjuan-VirtualBox:~/CPE212_SanJuan$ git commit -m "hello po"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
sanjuan@sanjuan-VirtualBox:~/CPE212_SanJuan$ git push
^C
sanjuan@sanjuan-VirtualBox:~/CPE212_SanJuan$ git push origin main
Everything up-to-date
sanjuan@sanjuan-VirtualBox:~/CPE212_SanJuan$ cat README.md
# CPE212_SanJuansanjuan@sanjuan-VirtualBox:~/CPE212_SanJuan$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. SSH (Secure Shell) is a secure protocol that allows users to remotely access and manage another computer through an encrypted connection, making it safe to execute commands, transfer files, and perform administrative tasks.

2. You'll know the public key has been successfully installed on the remote server when you can log in using SSH without being asked to enter a password.
3. So far, we have used Ansible commands to connect to remote servers, verify connectivity, execute commands remotely, gather system information, and automate administrative tasks.
4. The inventory file is important because it tells Ansible which remote servers to manage and provides the connection details needed to execute tasks on those servers.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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